Causal statistics for treatment models

2. Imperfect compliance

Ecole Normale Supérieure–PSL, L3 Economics and Cogmaster, 2025

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Today

We will explore a common situation when, in spite of randomization, there is selection into treatment. At the end of this lecture, you should:

- Formally distinguish assignment to treatement (Z) and actual treatment (T)
- Know what is imperfect compliance, how to measure it
- Understand the threat to identification it produces
- Know what you should NOT do to solve the problem
- Know what is the Intention to treat parameter, its meaning
- Know what is the Wald estimator, understand the intuition and know how to derive the proof

Starting example 1

Question: Does colonoscopy reduce the incidence of colorectal cancer?

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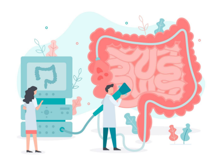
- Randomize an invitation to a free colonoscopy
- Measure everyone's colorectal cancer incidence 10 years later



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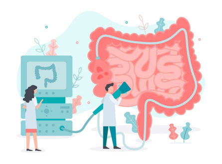

Of course:

Table 1. Sigmoidoscopy and colonoscopy screening trials

Trial	NordICC (1)	NORCCAP (2)	SCORE (3)	UKFSST (4)	PLCO (5)
Countries	Poland, Norway, Sweden	Norway	Italy	UK	U.S.
Screening period	2009 to 2014	1999 to 2001	1995 to 1999	1994 to 1999	1993 to 2001
Initial screening type	Colonoscopy	Sigmoidoscopy + FOBT	Sigmoidoscopy	Sigmoidoscopy	Sigmoidoscopy
Follow-up screening type	Biopsy	Colonoscopy, Biopsy	Colonoscopy	Colonoscopy	Second sigmoidoscopy (after 3/5 y)
Participants identification	Population registry	Population registry	Survey	Survey	Survey
Median Follow-up Years	10.0	10.9	10.5	11.2	11.9
Participants (N)	84,585	98,792	34,272	170,038	154,900
Range	55 to 64	50 to 64	55 to 64	55 to 64	55 to 74
Invitation ratio	0.33	0.21	0.50	0.34	0.50
Adherence rate	0.42	0.63	0.58	0.71	0.87

Starting example 2

Question: Does middle school principal/parents meetings improve student behavior?

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Of course:

TABLE 1
Take-up rate

	Treatment		Control	
	Volunteers	Non-vol's	Volunteers	Non-vol's
Initial workshop				
At least 1 debate	54.7	1.0	1.9	0.2
At least 2 debates	30.8	0.2	0.7	0.0
All 3 debates	13.5	0.1	0.7	0.0
Additional workshops	18.0	0.9	0.5	0.0

Note: All rates are expressed in percentage terms and are computed separately for each of the four groups.

Imperfect compliance

There is imperfect compliance whenever:

- not all subjects assigned to the treatment take the treatment

  or

- some subjects assigned to the control do take the treatment

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By design or de facto, this is an encouragement design

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Vocabulary

We now need to make a distinction between Assigned to treatment (Z) and Treated (T)

The problem when there is imperfect compliance

Principals meetings data

Open the Mallette data in your R Studio (it is data from the 2d paper, but on a subset where interesting features are more salient)

- "tirage" is assignment to treatment (invitation)
- "debat3" is actual treatment (attendance to meetings)

1 Compute the **take-up** of the treatment in the treatment and control groups

2 Compute the Δ compliance rate between the two groups, or net take-up

Take-up

Table 1: Proportion attending meetings

	Assigned treatment	Assigned control	Net take-up
	.547	**.019**	.528
N Obs.	539	431	

Take-up

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$$E(T|Z=1) = .547$$

$$E(T|Z=0) = .019$$

What is the problem?

Selectivity bias

- “behav-z-t1” is an index of good behavior at baseline (Trim 1)

  Are parents attending meetings selected compared to those not attending?

Selectivity bias

Table 2: Behavior at baseline

	All treated	All not treated	OLS of bl behavior on treated: Const.	Treated
	.051	**-.068**	-.068	.120
			(.038)	(.057)
p-value				.038
N Obs.	303	664		

Selectivity bias

$$E(X|T=1) \neq E(X|T=0)$$

Then it is likely that:

$$E(Y(0)|T=1) \neq E(Y(0)|T=0)$$

and

$$E(Y(1)|T=1) \neq E(Y(1)|T=0)$$

In general, T has not been randomized so no guarantee that $E(Y(0)|T=1) = E(Y(0)|T=0)$, etc.

What can we do about it?

Is the whole randomization effort ruined?

Awkward solutions

Table 3: OLS of behavior at **endline**

		Const.	Treated
Full sample		-.007	.134
		(.031)	(.060)
	p-value		.026
Excluding $(Z = 1, T = 0)$		-.030	.157
		(.040)	(.070)
	p-value		.025

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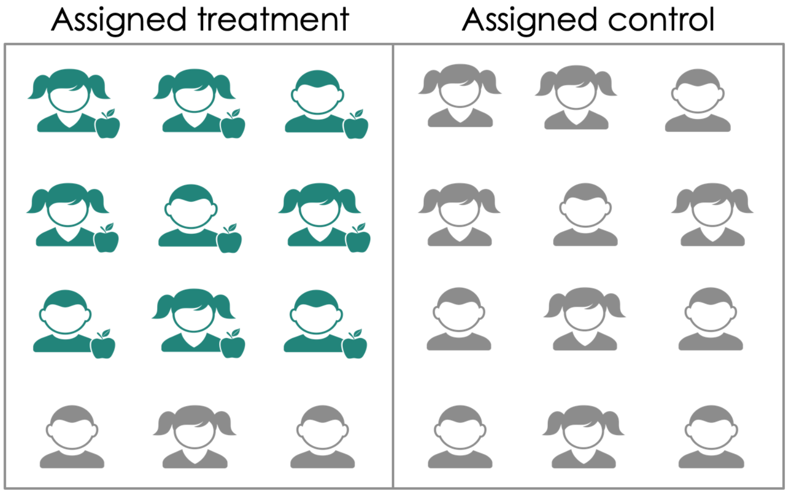
First line = naive estimator in presence of selection (cf lecture 1.)

The Intention to treat (ITT)

Which groups are comparable?



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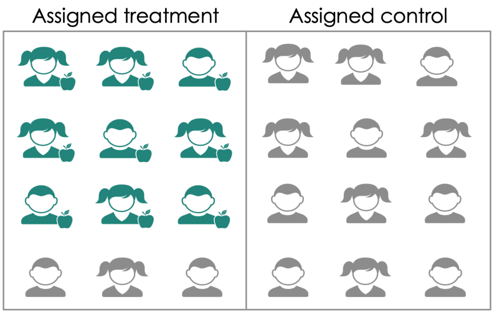
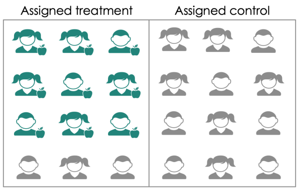


$$E(X|T=1) = .051 \text{ vs. } E(X|T=0) = -.068 \quad \text{(p-value=0.038)}$$

but

$$E(X|Z=1) = -.005 \text{ vs. } E(X|Z=0) = -.064 \quad \text{(p-value=0.362)}$$

Assume treatment effect is zero



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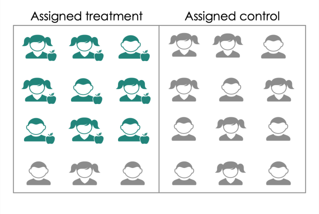

How much is $E(Y|Z=1) - E(Y|Z=0)$?

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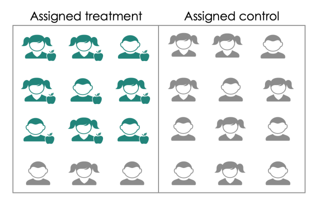

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If $Y(1) = Y(0)$ (zero treatment effect):

$$E(Y|Z=1) \quad = \quad E(Y|T=1, Z=1)P(T=1|Z=1) + E(Y|T=0, Z=1)P(T=0|Z=1)$$

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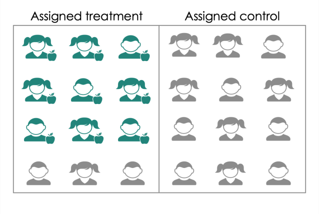

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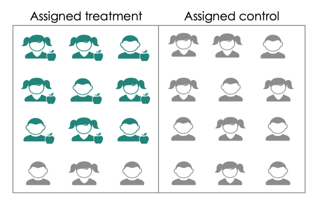

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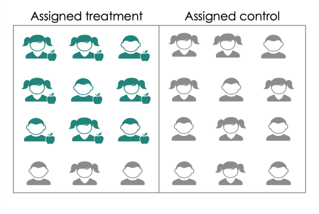

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&= E(Y(0)|Z=1) = E(Y(0)|Z=0) = E(Y|Z=0)
\end{aligned}
$$

Intention to treat parameter

Intention to treat

$$ITT = E(Y|Z = 1) - E(Y|Z = 0)$$

Intention to treat parameter

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Tests that there is an effect or not (and its sign)

Is the ITT an interesting parameter?

Intention to treat

- "behav-z-t3" is an index of good behavior at endline (Trim 3)

  Compute the ITT. Do the meetings have a causal effect?

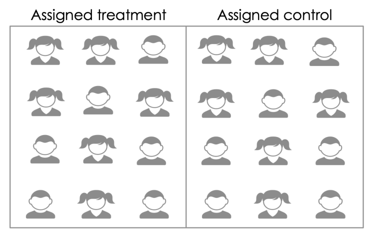
Intention to treat

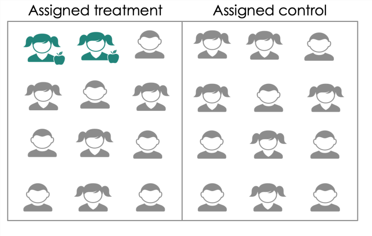
Table 4: Behavior at endline

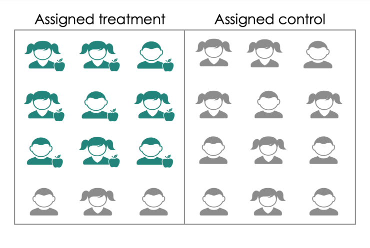
	Assigned treat.	Assigned contr.	OLS of behavior on Z:	
			Const.	Z
	.085	**-.028**	-.028	.113
			(.040)	(.058)
p-value				.053

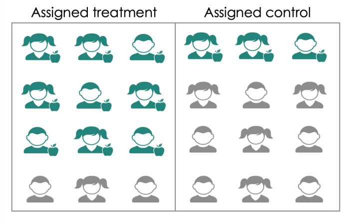
Recovering treatment effect: the Wald estimator

For a given effect of the treatment $\Delta = Y(1) - Y(0)$ (assume you know it), how does the ITT vary:









Build the ITT

Assume you know $\Delta_i = Y_i(1) - Y_i(0) = \Delta = 0.2$ (constant effect model)

Also simplify:

- $E(T|Z=1) = 0.5$ (it is in fact 0.547)
- $E(T|Z=0) = 0$ (it is in fact 0.019)

Intuitively, compute the ITT

Formally

$$E(Y|Z=0) \quad = \quad E(Y(0)|Z=0) = E(Y(0))$$

Formally

$$\begin{aligned} E(Y|Z=0) &= E(Y(0)|Z=0) = E(Y(0)) \\ E(Y|Z=1) &= E(Y|T=1, Z=1)P(T=1|Z=1) + E(Y|T=0, Z=1)P(T=0|Z=1) \end{aligned}$$

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Thus:

$$\text{ITT: } E(Y|Z=1) - E(Y|Z=0) =$$

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\end{aligned}
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Thus:

$$\text{ITT: } E(Y|Z=1) - E(Y|Z=0) = \Delta \times P(T=1|Z=1) = 0.2 \times 0.5 = 0.1$$

From ITT to Treatment effect

Conclusion: the ITT scales down the true effect by the take-up

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Let's prove it

Wald estimator

In the general case:

$$\begin{aligned} E(Y|Z=0) &= E(Y(0)|T=0, Z=0)P(T=0|Z=0) \\ &+ E(Y(1)|T=1, Z=0)P(T=1|Z=0) \end{aligned}$$

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\\
&= E(Y(0)|T=0,Z=0)P(T=0|Z=0) \\
&+ E(Y(0)|T=1,Z=0)P(T=1|Z=0) \\
&+ E(Y(1)-Y(0)|T=1,Z=0)P(T=1|Z=0)
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\\
&= E(Y(0)|Z=0) \\
&+ E(Y(1)-Y(0)|T=1,Z=0)P(T=1|Z=0) \\
\\
&= E(Y(0)) + \Delta P(T=1|Z=0)
\end{aligned}
$$

Wald estimator

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\begin{aligned}
E(Y|Z=1) &= E(Y(0)) + \Delta P(T=1|Z=1) \\
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where $P(T=1|Z=1)$ is take-up in the treatment group,
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$$\text{ITT: } E(Y|Z=1) - E(Y|Z=0) = \Delta \times [P(T=1|Z=1) - P(T=1|Z=0)]$$

Wald estimator

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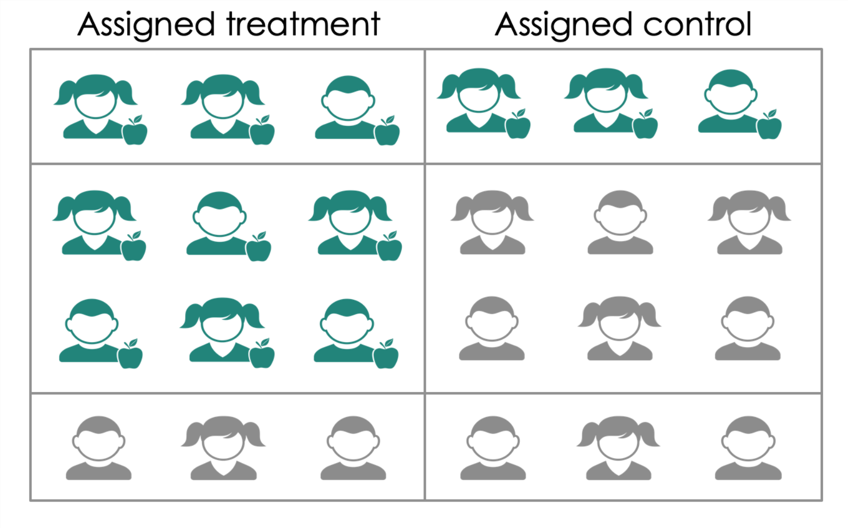
Wald estimator

$$
\Delta = \frac{E(Y|Z=1) - E(Y|Z=0)}{P(T=1|Z=1) - P(T=1|Z=0)}
$$

Wald estimator



Wald estimator

Get back to the data and compute the treatment effect of parental meetings on the behavioral index

Is there no problem?

Good news: imperfect compliance introduces selectivity into the treatment **but** it doesn't impair the identification of the treatment effect

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Still there are costs:

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Still there are costs:

1. If treatment effect is heterogeneous (Δ_i's vary): we don't identify the average treatment effect but an estimand called the LATE (we will get back to this in a later lecture)

2. ITT is equivalent to a regression of Y on Z. The coefficient $\hat{\beta}$ has a certain variance that depends on the variance of the residual and on the covariance matrix of $(1, Z_i)$

Is there no problem?

Good news: imperfect compliance introduces selectivity into the treatment **but** it doesn't impair the identification of the treatment effect

Still there are costs:

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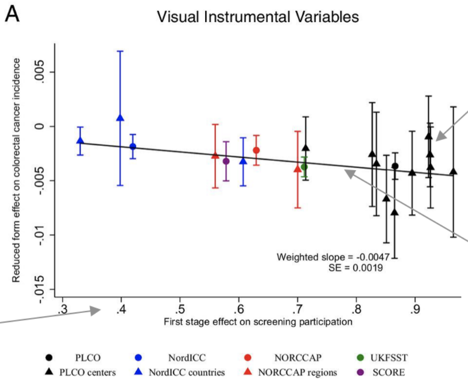
   If you decrease take-up, you mechanically decreases the ITT: when the coefficient $\hat{\beta}$ becomes too small relative to its variance, the effect is no longer "significant"
   So you can recover treatment effect, but with small statistical power if take-up is low

Back to colorectal cancer: ITT



Effet ITT sur l’incider
cancer colorectal

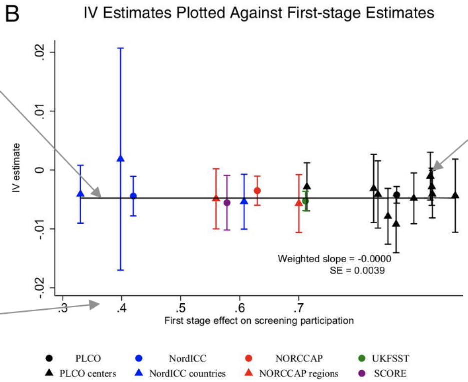
Pente de la relation
de compliers ~ ITT

Proportion de compliers

Figure 1 panel A, in Angrist et Hull (2023), “Instrumental variables methods reconcile intention-to-screen effects across pragmatic cancer screening trials”

Back to colorectal cancer: Wald estimator

Pente de la relation proportion de compliers ~ LATE



Effet LATE avec
de confiance à 9

Proportion de compliers

Figure 1 panel B, in Angrist et Hull (2023), "Instrumental variables methods reconcile intention-to-screen effects across pragmatic cancer screening trials"

Paper presentation: Castell L., Gurgand M., Imbert C. Tochev T., Take-up of Social Benefits: Experimental Evidence from France